ENEL 573 HW Assignment 4 (Fall 2026)

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Problem 1

Class B: | 1 0 | NET ID (14bit) | SUBNET ID (n bit) | HOST ID (16-n bit) |

2^n ≥ 950: 2^9 = 512 < 950, 2^10 = 1024 ≥ 950 → n = 10 bits subnet ID

HOST ID = 16 - n = 16 - 10 = 6 bits

Default mask: 11111111 11111111 00000000 00000000 = 255.255.0.0 (/16)

Subnet mask : 11111111 11111111 11111111 11000000

Ans: Subnet mask = /26 (prefix-length notation) = 255.255.255.192 (dot decimal notation)

2^10 = 1024 subnets, each has 2^6 = 64 addresses

Maximum 62 (2^6 - 2) hosts attachable to each subnet

Class B network address, e.g. 150.100.0.0 (10010110 = 150, 01100100 = 100)

Range address for subnet X, where X is 10 bits:

Starting address: 10010110 01100100 XXXXXXXX XX000000

Start address = subnet address

Last address   : 10010110 01100100 XXXXXXXX XX111111

X takes values from 0000000000 to 1111111111

Subnet #0:   150.100.0.0/26   to 150.100.0.63/26

Subnet #1:   150.100.0.64/26   to 150.100.0.127/26

Subnet #2:   150.100.0.128/26   to 150.100.0.191/26

Subnet #3:   150.100.0.192/26   to 150.100.0.255/26

...

Subnet #949: 150.100.237.64/26 to 150.100.237.127/26

...

Subnet #1023: 150.100.255.192/26 to 150.100.255.255/26

(Subnet #0 to #949 used for the 950 subnets, #950 to #1023 spare)

Problem 2

120 is between 0 and 127 → class A, default mask 255.0.0.0 (/8)

Subnet mask 255.255.192.0 = /18 → keep first 18 bits

Subnet mask bitwise ANDed with the IP address gives the subnet address:

IP address : 01111000 00001110 01111010 00010000 = 120.14.122.16

Subnet mask: 11111111 11111111 11000000 00000000 = 255.255.192.0

Subnet addr: 01111000 00001110 01000000 00000000 = 120.14.64.0

Ans: Subnet address = 120.14.64.0/18

Host id = remaining 32 - 18 = 14 bits of the IP address:

00000000 00000000 00111010 00010000 = 0.0.58.16

Ans: Host id = 0.0.58.16

Problem 3

Data = 1300 - 20 = 1280 bytes

Max data/fragment = 500 - 20 = 480 bytes (480/8 = 60 ✓)

1280 = 480 + 480 + 320

Frag	length	ID	flag	offset
1	=500	=X	=1	=0
2	=500	=X	=1	=480/8 = 60
3	=340	=X	=0	=960/8 = 120

Ans: Option 2 (offsets 0, 480/8, 960/8). Option 1 ignores the 20-byte header; option 3 subtracts it twice (460 is not a multiple of 8).

Problem 4

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Datagram OK only if all 10 fragments arrive.  p = 0.01
(a) P = (1 - p)^10 = (0.99)^10
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Ans: (a) P ≈ 0.9044

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(b) fragment lost only if both copies lost: p^2 = 0.0001
    P = (1 - p^2)^10 = (0.9999)^10
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Ans: (b) P ≈ 0.9990

Problem 5

Frames to B's MAC address: Yes, C's adapter processes them. On a broadcast LAN every frame reaches every adapter, and C's adapter checks the destination MAC. No, it does not pass the IP datagrams to C's network layer: the destination MAC is B's, not C's, so C's adapter discards the frame.

Frames to broadcast address FF-FF-FF-FF-FF-FF (ARP query): Yes, C's adapter processes them, and yes, it passes the contents up to C's network layer, since every adapter accepts the broadcast address. C then sees the target IP is B's, not its own, so it ignores the ARP query and does not reply (only B replies, by unicast to A).